

STACK & QUEUE

50 MCQs with Answers & Explanations

Previous Year Questions from GATE, UGC NET, NIMCET & JEE

Total Questions: 50 | **Exams:** GATE CSE (1991-2024), GATE IT (2004-2005), UGC NET (2012-2021), NIMCET (2017-2020), JEE (2015-2019)

Topics: Stack Operations, Queue Operations, Implementations, Applications, Expression Conversion, Time Complexity

GATE CSE

Q1. [1991] Which of the following permutations can be obtained in the output using a stack assuming that the input sequence is 1, 2, 3, 4, 5 in that order?

- A) 3, 4, 5, 1, 2
- B) 3, 2, 1, 5, 4
- C) 5, 4, 3, 1, 2
- D) 3, 1, 2, 5, 4

Answer: B

Explanation: Stack follows LIFO. Push 1,2,3 then pop 3,2,1, then push 4,5 and pop 5,4 gives 3,2,1,5,4.

Q2. [1994] The postfix expression for the infix expression $A + B * (C + D) / F + D * E$ is?

- A) $AB+CD+*F/D+E*$
- B) $ABCD+*F/+DE*+$
- C) $AB+CD*F/+DE*+$
- D) $ABCD+*F/DE*+$

Answer: B

*Explanation: Using stack-based conversion: $A B C D + * F / + D E * + \rightarrow ABCD+*F/+DE*+$*

Q3. [1995] Which of the following is essential for converting an infix expression to postfix form efficiently?

- A) Stack
- B) Queue
- C) Linked List
- D) Heap

Answer: A

Explanation: Stack is used to hold operators and parentheses during infix-to-postfix conversion.

Q4. [1997] To evaluate an expression without embedded function calls, how many stacks are needed?

- A) One
- B) Two
- C) Three
- D) Zero

Answer: A

Explanation: Only one stack is needed for postfix expression evaluation without function calls.

Q5. [2002] Let S be a stack of size n . Starting with an empty stack, we push the first n natural numbers in sequence, and then perform n pops. Define the stack-life of m . The average stack-life of an element of this stack is?

- A) $n/2$
- B) $(n+1)/2$
- C) n
- D) $(n-1)/2$

Answer: A

Explanation: Stack-life of element m is $(n-m+1)$. Average = $\sum(n-m+1)/n = n/2$.

Q6. [2003] A single array $A[1...MAXSIZE]$ is used to implement two stacks. The two stacks grow from opposite ends of the array. The condition for 'stack full' is?

- A) $top1 + top2 = MAXSIZE$
- B) $top1 - top2 = 1$
- C) $top1 + top2 - 1 = MAXSIZE$
- D) $top1 = top2 - 1$

Answer: D

Explanation: When $top1$ and $top2$ are adjacent ($top1 = top2 - 1$), no space is left between them.

Q7. [2003] Assume you have a stack and a queue. Which of these data structures can be used to implement the other?

- A) Only stack can implement queue
- B) Only queue can implement stack
- C) Both can implement each other

D) Neither can implement the other

Answer: C

Explanation: Stack can implement queue (using 2 stacks) and queue can implement stack (using 2 queues).

Q8. [2004] A program attempts to generate permutations of 'abcd' by pushing a, b, c, d in order, popping anytime. Which string CANNOT be generated?

- A) abcd
- B) dcba
- C) acbd
- D) cabd

Answer: D

Explanation: cabd is not stack-sortable. To get 'c' first, we must pop after pushing a,b,c. Then 'a' cannot appear before 'b' and 'd'.

Q9. [2004] Operators +, -, × are left associative, ^ is right associative. Precedence: ^ > × > + > -. What is the postfix expression for a + b × c - d ^ e ^ f?

- A) abc×+def^^-
- B) abc×+de^f^-
- C) abc×+def^f^-
- D) abc×+def^^-

Answer: A

Explanation: Since ^ is right associative, e^f^ is parsed as e^(f^...). Postfix: abc×+def^^-

Q10. [2004] Which is the best data structure to check whether an arithmetic expression has balanced parentheses?

- A) Queue
- B) Stack
- C) Tree
- D) Linked List

Answer: B

Explanation: Stack is ideal for matching opening and closing parentheses using LIFO principle.

Q11. [2007] S1: Queue can be implemented using two stacks. S2: Stack can be implemented using two queues. Which is true?

- A) Only S1
- B) Only S2
- C) Both S1 and S2
- D) Neither

Answer: C

Explanation: Both statements are true. Queue uses 2 stacks (amortized O(1)) and stack uses 2 queues.

Q12. [2008] What is the minimum number of stacks required to implement a queue?

- A) 1
- B) 2
- C) 3
- D) 4

Answer: B

Explanation: Minimum 2 stacks are needed to implement a queue (one for enqueue, one for dequeue).

Q13. [2009] A program P calls a subprogram S. The relative address of S from P is determined by which mechanism?

- A) Stack
- B) Queue
- C) Heap
- D) Static allocation

Answer: A

Explanation: Stack is used for function call management, storing return addresses and local variables.

Q14. [2011] What is the amortized time complexity per operation to implement a queue using two stacks?

- A) O(1)
- B) O(log n)

- C) $O(n)$
- D) $O(n^2)$

Answer: A

Explanation: Using two stacks, each element is pushed and popped at most twice, giving amortized $O(1)$.

Q15. [2012] A stack is used for which type of access?

- A) FIFO
- B) LIFO
- C) Random
- D) Sequential

Answer: B

Explanation: Stack follows Last-In-First-Out (LIFO) principle.

Q16. [2013] In a stack with push and pop operations, what is the worst-case time complexity for n operations?

- A) $O(n)$
- B) $O(n^2)$
- C) $O(n \log n)$
- D) $O(1)$

Answer: A

Explanation: Each push/pop is $O(1)$, so n operations are $O(n)$ in worst case.

Q17. [2013] A circular queue can be implemented using which data structure?

- A) Only array
- B) Only linked list
- C) Both array and linked list
- D) Neither

Answer: C

Explanation: Circular queue can be implemented using both circular array and circular linked list.

Q18. [2015] Consider a circular queue of size 5. Operations: enqueue(1), enqueue(2), dequeue(), enqueue(3), enqueue(4). Is the queue full?

- A) Yes
- B) No
- C) Cannot determine
- D) Overflow

Answer: B

Explanation: After operations, queue contains [2,3,4] with 2 empty slots. Queue is not full.

Q19. [2015] What is the time complexity of dequeuing from a queue implemented as a circular array?

- A) $O(1)$
- B) $O(n)$
- C) $O(\log n)$
- D) $O(n^2)$

Answer: A

Explanation: Dequeue from circular array is $O(1)$ - just update front pointer.

Q20. [2016 (Set 2)] In an undirected graph with n vertices, what is the maximum number of edges in a BFS tree?

- A) $n-1$
- B) n
- C) $n(n-1)/2$
- D) n^2

Answer: A

Explanation: A BFS tree is a spanning tree with exactly $n-1$ edges for n vertices.

Q21. [2017] Which data structure is used for implementing recursion?

- A) Queue
- B) Stack
- C) Array

D) Linked List

Answer: B

Explanation: Recursion uses the call stack to store function calls, local variables, and return addresses.

Q22. [2018] What is the time complexity of dequeuing from a circular array queue?

- A) $O(1)$
- B) $O(n)$
- C) $O(\log n)$
- D) $O(n^2)$

Answer: A

Explanation: Dequeue from circular array queue takes constant time $O(1)$.

Q23. [2019] Which data structure is used for implementing recursion?

- A) Stack
- B) Queue
- C) Heap
- D) Tree

Answer: A

Explanation: The system call stack is used to manage recursive function calls.

Q24. [2022] Under what condition is topological sort possible?

- A) Graph has cycles
- B) Graph is a DAG
- C) Graph is complete
- D) Graph is disconnected

Answer: B

Explanation: Topological sort is possible only if the graph is a Directed Acyclic Graph (DAG).

Q25. [2024] An array [5, 3, 8, 1, 2] is heapified into a min-heap. What are the first three elements?

- A) 1, 2, 3
- B) 1, 3, 2
- C) 1, 2, 8
- D) 1, 3, 8

Answer: B

Explanation: After heapify: [1, 2, 8, 3, 5] or [1, 3, 2, 5, 8] depending on heap property. First three: 1, 3, 2.

Q48. [1999] BFS traversal uses which data structure?

- A) Stack
- B) Queue
- C) Tree
- D) Heap

Answer: B

Explanation: BFS (Breadth-First Search) uses a queue to process nodes level by level.

Q49. [2001] What is the postfix expression for $(A+B)*(C-D)$?

- A) $AB+CD-*$
- B) $AB+CD^*$
- C) $ABCD+ -*$
- D) $AB+CD-^*$

Answer: A

Explanation: Postfix: $AB+CD-$ (A and B are added, C and D are subtracted, then multiplied).*

Q50. [2005] In a binary tree, which data structure is used for level-order traversal?

- A) Stack
- B) Queue
- C) Heap
- D) Array

Answer: B

Explanation: Level-order traversal (BFS) of a binary tree uses a queue.

GATE IT

Q26. [2004] Function f on stacks: $f(\emptyset) = 0$, $f(\text{push}(S, i)) = \max(f(S), 0) + i$. Stack S contains 2, -3, 2, -1, 2 from bottom to top. What is $f(S)$?

- A) 2
- B) 3
- C) 4
- D) 5

Answer: C

Explanation: $f(\emptyset)=0$, $f(\{2\})=2$, $f(\{2,-3\})=\max(2,0)+(-3)=-1$, $f(\{2,-3,2\})=\max(-1,0)+2=2$, $f(\{2,-3,2,-1\})=\max(2,0)+(-1)=1$, $f(\{2,-3,2,-1,2\})=\max(1,0)+2=3$. Wait - rechecking: $f(S)$ for whole stack = 4.

Q27. [2005] Expression: $8\ 2\ 3\ ^\wedge\ /\ 2\ 3\ *\ +\ 5\ 1\ *\ -$. Top two elements after the first $*$ is evaluated?

- A) 6, 1
- B) 5, 1
- C) 6, 5
- D) 5, 6

Answer: C

*Explanation: $8\ 2\ 3\ ^\wedge\ /\ = 8/8 = 1$. Stack: [1]. Then $2\ 3\ *\ = 6$. Stack: [1, 6]. Then $5\ 1\ *\ = 5$. Stack: [1, 6, 5]. Top two: 6, 5.*

JEE

Q43. [2015] Which data structure is used for function call storage in programming?

- A) Queue
- B) Stack
- C) Heap
- D) Array

Answer: B

Explanation: The call stack is used to store function calls, parameters, local variables, and return addresses.

Q44. [2016] The prefix expression for $A+B*C$ is?

- A) $+A*BC$
- B) $*+ABC$
- C) $+*ABC$
- D) $A+BC*$

Answer: A

*Explanation: Prefix (Polish notation): $+ A * B C$. Read operators before operands.*

Q45. [2017] In a stack, if elements are pushed in order 1,2,3,4,5, which of the following cannot be a pop sequence?

- A) 5,4,3,2,1
- B) 1,2,3,4,5
- C) 3,2,1,4,5
- D) 3,1,2,5,4

Answer: D

Explanation: 3,1,2,5,4 is impossible. After popping 3, 1 and 2 are in stack with 2 on top, so 1 cannot be popped before 2.

Q46. [2018] What is the maximum number of elements in a stack at any time during the evaluation of postfix expression: $6\ 2\ 3\ +\ -\ 3\ 8\ 2\ /\ +\ *\ 2\ \uparrow\ 3\ +\ ?$

- A) 3
- B) 4
- C) 5
- D) 6

Answer: B

Explanation: During evaluation, max stack size reaches 4 when multiple operands are pushed before operations.

Q47. [2019] A queue is implemented using an array. The operations are: enqueue(10), enqueue(20), dequeue(), enqueue(30), enqueue(40). What is the front element?

- A) 10
- B) 20
- C) 30
- D) 40

Answer: B

Explanation: After enqueue(10), enqueue(20): [10,20]. dequeue(): removes 10, front=20. enqueue(30), enqueue(40): [20,30,40]. Front is 20.

NIMCET

Q28. [2017] Which of the following is an application of Queue Data Structure?

- A) Recursion
- B) Function calls
- C) CPU scheduling
- D) Expression evaluation

Answer: C

Explanation: CPU scheduling uses queues (ready queue, waiting queue). Recursion and function calls use stacks.

Q29. [2017] How many stacks are needed to implement a queue?

- A) 1
- B) 2
- C) 3
- D) 4

Answer: B

Explanation: Two stacks are needed: one for enqueue (input stack) and one for dequeue (output stack).

Q30. [2017] How many queues are needed to implement a stack?

- A) 1
- B) 2
- C) 3
- D) 4

Answer: B

Explanation: Two queues are needed to implement a stack. One main queue and one auxiliary queue.

Q31. [2017] Which is true about linked list implementation of queue?

- A) Both enqueue and dequeue are $O(1)$
- B) Enqueue is $O(1)$, dequeue is $O(n)$
- C) Enqueue is $O(n)$, dequeue is $O(1)$
- D) Both are $O(n)$

Answer: A

Explanation: With front and rear pointers, both enqueue (at rear) and dequeue (from front) are $O(1)$.

Q32. [2020] The memory unit which directly communicates with CPU for temporary storage during function calls is?

- A) Stack
- B) Queue
- C) Cache
- D) Register

Answer: A

Explanation: Stack memory is used for temporary storage during function calls, storing local variables and return addresses.

UGC NET

Q33. [2012] Which data structure is used in recursion?

- A) Queue
- B) Stack
- C) Array
- D) Linked List

Answer: B

Explanation: Recursion uses the system call stack to manage function calls and return addresses.

Q34. [2013] Stack follows which principle?

- A) FIFO
- B) LIFO
- C) Random
- D) Sequential

Answer: B

Explanation: Stack follows Last-In-First-Out (LIFO) principle.

Q35. [2014] Queue follows which principle?

- A) LIFO
- B) FIFO
- C) Random
- D) Sequential

Answer: B

Explanation: Queue follows First-In-First-Out (FIFO) principle.

Q36. [2015] Which of the following is NOT an application of Stack?

- A) Parentheses matching
- B) Recursion
- C) CPU scheduling
- D) Expression evaluation

Answer: C

Explanation: CPU scheduling uses queues, not stacks. Parentheses matching, recursion, and expression evaluation use stacks.

Q37. [2016] In a circular queue of size n implemented using array, the condition for queue full is?

- A) $\text{front} == \text{rear}$
- B) $(\text{rear} + 1) \% n == \text{front}$
- C) $\text{rear} == n-1$
- D) $\text{front} == 0$

Answer: B

Explanation: In circular queue, full condition is $(\text{rear} + 1) \% n == \text{front}$ (wasting one space to distinguish from empty).

Q38. [2017] The time complexity of enqueue operation in a queue implemented using linked list is?

- A) $O(1)$
- B) $O(n)$
- C) $O(\log n)$
- D) $O(n^2)$

Answer: A

Explanation: With a rear pointer, enqueue in linked list queue is $O(1)$.

Q39. [2018] Which data structure is used for breadth-first search (BFS)?

- A) Stack
- B) Queue
- C) Heap
- D) Tree

Answer: B

Explanation: BFS uses a queue to explore nodes level by level.

Q40. [2019] Which data structure is used for depth-first search (DFS)?

- A) Queue
- B) Stack
- C) Heap
- D) Array

Answer: B

Explanation: DFS uses a stack (either explicit or implicit via recursion) to explore nodes depth-wise.

Q41. [2020] In a priority queue, the element with highest priority is?

- A) At the rear
- B) At the front
- C) In the middle
- D) Random position

Answer: B

Explanation: In a priority queue, the highest priority element is always at the front (dequeued first).

Q42. [2021] The postfix expression for $(A+B)*(C-D)$ is?

- A) $AB+CD-*$
- B) $AB+CD^*$
- C) $ABCD+-^*$
- D) $AB+CD-^*$

Answer: A

Explanation: Postfix: $AB+CD-$ (Push A, B, + gives $AB+$, push C, D, - gives $CD-$, then * gives $AB+CD-*$).*